

Yemen

Architecture Initiative



PROJECT DOCUMENTATION — COMPLETE EDITION

23

Architectural
characters

13

Kingdom
archives

8

YBC
chapters

51

Scholarly
sources

PWA Ready

WCAG AAA

UNESCO Aligned

ALIPH Protocol

Offline First

Table of Contents

1.	Executive Summary	3
2.	Platform Overview	4
3.	Five Sectors	5
4.	23 Architectural Characters	7
5.	Yemen Building Code (YBC)	10
6.	13 Kingdom Archives	13
7.	Scholarly Foundation — 51 Sources	16
8.	Design System	18
9.	Technical Architecture	19
10.	Analytics Strategy	21
11.	PWA & Offline Architecture	22
12.	SEO & Browser Introduction	23
13.	Deployment Guide	24
14.	Global Partners	25
15.	Roadmap	26

Executive Summary

The Yemen Architecture Initiative (YAI) is a Progressive Web Application and global knowledge platform dedicated to documenting, preserving, and communicating Yemen's extraordinary architectural heritage. Built in response to the urgent convergence of conflict damage, generational craft loss, and global unfamiliarity with Yemen's built environment, YAI delivers a comprehensive platform across five content sectors, serving architects, government bodies, students, the global public, and the Yemeni diaspora.

Yemen hosts three UNESCO World Heritage Sites — the Old City of Sana'a (1986), the Old Walled City of Shibam (1982), and the Historic Town of Zabid (1993) — plus the Landmarks of the Ancient Kingdom of Saba at Ma'rib (2023). These four inscriptions represent only a fraction of Yemen's architectural significance: 22 governorates each carry distinct vernacular traditions shaped by geology, climate, tribal structure, and 3,000 years of civilisational continuity from the Sabaeen temples of Ma'rib to the living tower houses of Sana'a's Old City.

YAI's immediate mission is threefold: to document what exists before it is lost; to codify construction standards that allow authentic restoration and reconstruction through the Yemen Building Code Heritage Module; and to make this knowledge globally accessible through a platform that works offline in low-connectivity fieldwork conditions — including inside Yemen itself.

KEY PLATFORM METRICS

Metric	Value	Notes
Architectural characters	23	22 governorates + Sheba Kingdom
Kingdom archives	13	From Awsan (800 BCE) to Ottoman (1918)
YBC Heritage chapters	8	Modelled on Saudi SBC 902
Scholarly sources	51	English, French, German, Italian, journals
Target audiences	4+	Architects, government, students, public, diaspora
UNESCO sites included	4	Sana'a, Shibam, Zabid, Sheba/Ma'rib
Languages	2	Arabic (RTL) and English (LTR)
PWA features	Full	Offline, installable, push notifications, background sync
WCAG compliance	AAA	Keyboard navigable, screen reader tested

Platform Overview

YAI is structured around five content sectors, each serving a distinct audience with a distinct content depth and purpose. The platform is built as a Progressive Web Application — accessible via browser worldwide, installable as a native-feeling app, and functional offline in the field.

Platform Architecture

File	Page	Primary audience	Core content
index.html	Homepage	All visitors	Hero, sectors, Sheba feature, map, statistics
map.html	Interactive Map	Architects, researchers	23 character zones, filterable, click-to-detail
students.html	Students sector	Students, educators	Learning hub, research library, toolkits
code.html	YBC sector	Architects, government	8 building code chapters, downloadable PDFs
details.html	Details sector	Architects, researchers	Character cards, materials, 3D documentation
general.html	General sector	Public, diaspora	22 city guides, diaspora portal, news
archives.html	Archives sector	Researchers, public	13 kingdom profiles, 2,700-year timeline
offline.html	Offline fallback	All (offline)	Cached content navigator, retry mechanism
sw.js	Service Worker	— (background)	Workbox caching, background sync, push
manifest.json	PWA Manifest	— (browser)	Install configuration, shortcuts, icons

Five Sectors

Sector 01

Students

Educational resources for architecture students, university programmes, schools, and young people.

- Learning hub — structured courses (beginner / intermediate / advanced)
- University resources — lecture decks, studio briefs, measured drawing exercises
- Visual dictionary — every architectural term illustrated in Arabic and English
- Research library — 51 sources across 5 languages, with abstracts and open-access links
- Classroom toolkit — K-12 teacher packs, colouring pages, model-building guides
- Annual global design competition — using Yemen's 23 characters as studio brief

Sector 02

Code (YBC)

Yemen Building Code Heritage Module — technical standards for construction, restoration, and reconstruction.

- YBC-H00 — Overview & legal framework, heritage classification grades A–D
- YBC-H01 — Mudbrick (libn) — mix ratios, height limits, annual render protocols
- YBC-H02 — Fired brick & gypsum — Sana'a tradition, qamaria fabrication standards
- YBC-H03 — Coral stone & timber — Tihama rawshan screens, moisture management
- YBC-H04 — Rammed earth (zabur) — Sa'dah northern tradition, seismic performance
- YBC-H05 — Seismic standards — highland tower house retrofit, reversible intervention
- YBC-H06 — Flood resilience — wadi-adjacent platforms, Shibam special protocol
- YBC-H07 — Post-conflict reconstruction — CDC-1 to CDC-5 damage classification

Sector 03

Details

Deep technical documentation for architects and researchers.

- 23 character cards — history, climate response, materials, typology drawings
- Each card includes 3 design typologies: traditional, transitional, contemporary
- Material library — libn, fired brick, coral, basalt, gypsum, reed, alabaster
- Element catalogue — qamaria, mafraj, rawshan, birka, samsarah defined and measured
- 3D documentation — partnership with Iconem and CEFAS digital scans
- Builder knowledge archive — video oral histories from master craftsmen (usta)

Sector 04

General

Cultural storytelling for the global public and Yemeni diaspora.

- Yemen's architectural story — narrative from Sheba Kingdom to today
- 22 city guides — accessible profiles for non-specialist readers
- Diaspora portal — community memory submissions, "my city" personal stories
- Heritage news — conservation updates, UNESCO and ALIPH developments
- Global partners directory — UNESCO, ALIPH, CEFAS, DAI, MAIRY, AIYS
- Donate & support — links to verified heritage conservation organisations

13 kingdoms and dynasties across 2,700 years — pre-Islamic to Ottoman.

- 6 pre-Islamic kingdoms — Awsan, Saba, Ma'in, Qataban, Hadramaut, Himyar
- 7 Islamic dynasties — Ziyadid, Sulayhid (Queen Arwa), Rasulid, Tahirid, Ottoman I, Qasimi, Ottoman II
- Interactive timeline rail — scrollable chronological navigation
- Each kingdom: architectural profile, key sites, scholarship, research gaps
- Sheba Kingdom as foundational layer — appears across all 5 sectors
- Cross-links to modern governorate characters showing civilisational continuity

23 Architectural Characters

Every governorate of Yemen carries a distinct architectural DNA shaped by its geology, climate, tribal structure, and trade connections. YAI documents 23 characters — 22 governorate zones plus the Sheba Kingdom as a foundational layer.

★ SHEBA KINGDOM — FOUNDATIONAL CHARACTER

Ma'rib · Al Jawf · Shabwah · c. 900 BCE –
275 CE · UNESCO 2023

HIGHLAND STONE TRADITION

#	Character	Governorate	Period	Signature element
01	Sana'a Capital	Amanat al-Asimah	Pre-Islamic–present	Qamaria · mafraj · fired brick
02	Sana'a Governorate	Sana'a	Medieval–present	Highland village · stone
04	Amran	Amran	Medieval–present	Walled town · defensive gate
06	Ad Dali'	Ad Dali'	Ottoman–present	Ottoman hill town · stone
07	Al Bayda	Al Bayda	Medieval–present	Central transitional
11	Al Mahwit	Al Mahwit	Medieval–present	Cliffside terraced village
13	Dhamar	Dhamar	Medieval–present	Volcanic basalt stone
15	Hajjah	Hajjah	Medieval–present	Cliffside · defensive · birka
16	Ibb	Ibb	Rasulid–present	Green highland · medieval stone
19	Raymah	Raymah	Pre-modern–present	Isolated terraced village
22	Ta'izz	Ta'izz	Rasulid–present	Fortress city · golden age

DESERT & EARTH TRADITION

#	Character	Governorate	Material	Signature element
09	Al Jawf	Al Jawf	Mudbrick · stone	Desert oasis · walled city
20	Sa'dah	Sa'dah	Rammed earth (zabur)	Zabur courses · walled city

COASTAL & MARINE TRADITION

#	Character	Governorate	Material	Signature element
03	Aden	Aden	Coral · colonial brick	Colonial hybrid · port
05	Abyan	Abyan	Clay · coral	Lowland courtyard house
08	Al Hudaydah	Al Hudaydah	Coral stone · timber	Rawshan screen · Zabid
10	Al Mahrah	Al Mahrah	Coral · mud	Frankincense route coastal

#	Character	Governorate	Material	Signature element
17	Lahij	Lahij	Coral · clay	Sultanate courtyard palace

ANCIENT & PRE-ISLAMIC TRADITION

#	Character	Governorate	Period	Signature element
14	Hadramawt	Hadramawt	Pre-Islamic–present	Mudbrick skyscrapers · Shibam · Tarim
18	Ma'rib	Ma'rib	Sabaeen–present	Awwam Temple · Great Dam · UNESCO
21	Shabwah	Shabwah	Hadramite–present	Palatial mudbrick · frankincense

ISLAND VERNACULAR

#	Character	Governorate	Material	Signature element
12	Socotra	Socotra Archipelago	Cave stone · endemic timber	Cave dwelling · dragon blood tree

Yemen Building Code (YBC)

The YBC Heritage Module is the technical regulatory framework for construction, restoration, and reconstruction across Yemen's heritage zones. Modelled on Saudi Arabia's SBC 902 (launched March 2025), it is adapted specifically to Yemen's four primary material traditions, geological conditions, and post-conflict reconstruction context.

Comparison with Saudi SBC 902

Dimension	Saudi SBC 902	Yemen YBC Heritage Module
Primary driver	National pride + Vision 2030	Cultural rescue + post-conflict reconstruction
Urgency	Preventive	Emergency + preventive
Living builders	Some knowledge remaining	Rapidly disappearing — documented now
Special chapter	N/A	YBC-H07: Conflict damage reconstruction
Material focus	Stone · mudbrick (Najd)	Libn · fired brick · coral · zabur
International partners	Mainly internal	UNESCO · ALIPH · CEFAS · Iconem
Output emphasis	New buildings reflect heritage	Restore existing + guide new

Chapter Summary

Code	Chapter	Tradition	Key spec	Govs	Status
H00	Overview & Framework	All	Grades A–D, 45-day review	All 22	Under review
H01	Mudbrick (Libn)	Hadramawt/desert	Max 8 storeys, annual render	5 govs	Draft
H02	Fired Brick & Gypsum	Sana'a highland	No Portland cement on exterior	5 govs	Draft
H03	Coral Stone & Timber	Tihama coastal	Lime mortar only, rawshan spec	4 govs	Draft
H04	Rammed Earth (Zabur)	Northern highland	300mm course height, seismic	3 govs	Draft
H05	Seismic Standards	Highland tower	Zones 1–3, reversible retrofit	6 govs	Under review
H06	Flood Resilience	Wadi-adjacent	Platform inspection quarterly	5 govs	Draft
H07	Post-Conflict	All	CDC-1–5, ALIPH alignment	All 22	Draft

13 Kingdom Archives

The Archives sector documents 13 kingdoms and dynasties across 2,700 years of Yemeni history. Each archive entry includes an architectural profile, key sites, primary scholarship, and direct cross-links to the modern governorate characters whose building traditions descend from each kingdom.

Pre-Islamic Kingdoms (c. 800 BCE – 525 CE)

Kingdom	Dates	Capital	Architectural legacy	Primary scholarship
Awsan	c. 800–500 BCE	Wadi Markha	Early temple complex, ashlar quoins	YAI original research
Saba (Sheba)	c. 900–275 BCE	Marib · Sirwah	Awwam Temple, Great Dam, ashlar Doric colonnade	Dr. Asma MYRI · CEFAS
Ma'in	c. 400–100 BCE	Qarnaw (Al Jawf)	Baraqish city wall (intact), caravan fort	MAIRY Italian mission
Qataban	c. 400 BCE–1st c.	Timna (Harib)	Most complete pre-Islamic urban plan	Loireto 2011 (Naples)
Hadramaut	c. 400 BCE–3rd c.	Shabwa	Palatial mudbrick — direct antecedent to Sana'a	Bruton Shill · CEFAS
Himyar	110 BCE–525 CE	Zafar · Sana'a	Raidan palace, early Sana'a foundations	Foster · Daum 1988

Islamic Dynasties (819 CE – 1918 CE)

Dynasty	Dates	Capital	Architectural legacy	Significance
Ziyadid	819–1018 CE	Zabid	Established Zabid — coral city	Founded UNESCO city
Sulayhid	1047–1138 CE	Jiblah	Queen Arwa's mosque & palace	Only ruling queen in Yemen
Rasulid	1229–1454 CE	Ta'izz · Zabid	Finest carved stucco in Arabia	Architectural golden age
Tahirid	1454–1526 CE	Rada'	Great Mosque of Rada'	Post-Rasulid patronage
Ottoman I	1538–1635 CE	Sana'a	Pencil minarets, garrison buildings	Brief but visible imprint
Qasimi	1597–1962 CE	Sana'a · Sa'dah	Tower house refinement, zabur peak	Longest dynasty (365yr)
Ottoman II	1849–1918 CE	Sana'a · Hodeidah	Hukumat buildings, Hodeidah port	Admin architecture

Scholarly Foundation — 51 Sources

YAI's content is grounded in 51 peer-reviewed and scholarly sources across five languages. The research library is accessible through the Students sector and provides the evidentiary basis for all character cards, YBC chapter specifications, and kingdom archive entries.

English

18 sources

- Damluji, S.S. — The Architecture of Yemen and Its Reconstruction (2017)
- Marchand, T.H.J. (ed.) — Architectural Heritage of Yemen (2017)
- Serjeant, R.B. & Lewcock, R. — San'a': An Arabian Islamic City (1983)
- Marchand, T.H.J. — Minaret Building and Apprenticeship in Yemen (2001)
- De Maigret, A. — Arabia Felix: Archaeological History of Yemen (2002)
- Daum, W. (ed.) — Yemen (English edition, 1988)

Français

12 sources

- Hirschi, S. & M. — L'Architecture du Yémen du Nord (1983)
- Bonnenfant, P. — Zabid au Yémen: Archéologie du vivant (2004)
- Bonnenfant, P. — Les maisons de Zabid (2008)
- Chelhod, J. — Culture et institutions du Yémen (1985)
- Chroniques Yéménites (CEFAS) — ongoing journal series (open access)

Deutsch

8 sources

- Daum, W. (Hrsg.) — Jemen (1988, ISBN 3-7016-2251-5)
- Gerlach, I. (Hrsg.) — 25 Jahre Ausgrabungen im Jemen (DAI, 2003)
- Finster, B. — Die Architektur der Rasuliden (in Daum, 1988)
- Shibam: Leben in Lehmtürmen (Reichert Verlag)

Italiano

7 sources

- Nicoletti, M. — Architettura e paesaggio nello Yemen del Nord (1992)
- Loreto, R. — L'architettura domestica e i palazzi reali (2011)
- MAIRY — Missione Archeologica Italiana in Yemen (1983–2010)

Journals

6 sources

- Proceedings of the Seminar for Arabian Studies (annual)
- Arabian Archaeology and Epigraphy (Wiley)
- Sabaean Studies — *Dirāsāt Saba'iyya* (Naples, 2005)
- Yemen Update (AIYS biannual)
- British-Yemeni Society Journal (annual)

Design System

Generated by the UI-UX-Pro-Max skill and refined with editorial overrides for the specific requirements of a bilingual Arabic/English cultural heritage platform.

Colour Tokens

Token	Hex	RGB	Usage
--ink	#0F172A	15, 23, 42	Headings, nav background, hero
--stone	#334155	51, 65, 85	Subheadings, body emphasis
--muted	#64748B	100, 116, 139	Captions, placeholders, tertiary text
--sky	#0369A1	3, 105, 161	CTA buttons, links, Students accent
--parchment	#F8FAFC	248, 250, 252	Page background, card backgrounds
--sand	#C4A77D	196, 167, 125	Yemen earth tone, nav accent, Sheba
--gold	#B45309	180, 83, 9	Sheba Kingdom, Details accent, gold CTA
--c-students	#0369A1	3, 105, 161	Students sector accent
--c-code	#15803D	21, 128, 61	Code (YBC) sector accent
--c-details	#B45309	180, 83, 9	Details sector accent
--c-general	#9F1239	159, 18, 57	General sector accent
--c-archives	#7C3AED	124, 58, 237	Archives sector accent

Typography

Role	Font family	Weights	Language	Usage
Display/heading	Cormorant Garamond	400, 500, 600	English	Hero text, section titles, character names
Body	DM Sans	300, 400, 500	English	All body copy, UI labels, descriptions
Technical	Fira Code	400, 500	EN (monospace)	YBC specifications, code snippets, stats
AR Display	Noto Naskh Arabic	400, 500, 600, 700	Arabic (RTL)	Arabic headings, Sheba section
AR Body	Noto Sans Arabic	300, 400, 500, 700	Arabic (RTL)	Arabic body text, navigation

Technical Architecture

Technology Stack

Layer	Technology	Version	Purpose
Base	HTML / CSS / JavaScript	ES2022+	Zero-dependency — maximum longevity and accessibility
PWA	Workbox service worker	7.0.0	CacheFirst for YBC PDFs, StaleWhileRevalidate for cards
Fonts	Google Fonts	Variable	Cormorant · DM Sans · Fira Code · Noto Arabic family
Analytics	Plausible Analytics	Latest	Privacy-first, no cookies, GDPR compliant, \$9/mo
PDF export	ReportLab (Python)	4.x	YBC chapter PDF generation
Push notif.	Firebase Cloud Messaging	v9	Heritage news push notifications
CI/CD	GitHub Actions	v4	Lighthouse audit + Vercel deploy on main branch push
Hosting	Vercel (recommended)	Latest	Global edge, Bahrain + Dubai nodes for ME speed
Monitoring	Vercel Analytics	Latest	Core Web Vitals monitoring

Workbox Caching Strategies

Content type	Strategy	Max age	Reason
Character cards & sector pages	StaleWhileRevalidate	7 days	Always fresh but instant from cache
YBC PDF chapters	CacheFirst	60 days	Offline priority — field use without connectivity
Map tiles	CacheFirst	30 days	Offline mapping in Yemen
Images	CacheFirst	30 days	Photography loads instantly
Google Fonts	CacheFirst	365 days	Typography never re-downloads
Navigation pages	NetworkFirst	30 day fallback	Always latest content when online
Plausible analytics	NetworkOnly	N/A	Analytics must never be cached

Analytics Strategy

YAI uses a three-layer analytics strategy built on Plausible Analytics — a privacy-first, GDPR-compliant platform that requires no cookie consent banner, works in Arabic-speaking markets, and costs \$9/month.

Layer	What is tracked	Tool	Frequency
1 — Traffic	Total visitors, unique, geography, device, browser, PWA install page views	Plausible page viewer	Real-time
2 — Roles	Architect / student / government / public / researcher	Plausible first visit role prompt	Per session
3 — Engagement	Sector entry, YBC chapter downloads, 3D model opens, site event clicks, digital submissions	Plausible events	Reported on 15 mins

Role segmentation is the key analytical innovation — a one-question onboarding prompt on first visit sets a role tag in localStorage. All subsequent Plausible events are tagged with this role, enabling breakdown of "which sector do architects use most?" vs "which pages do students visit first?" without any personally identifiable information.

PWA & Offline Architecture

Feature	Implementation	YAI benefit
Installable	Web App Manifest (manifest.json)	Add to home screen without app store
Offline-first	Workbox service worker (sw.js)	Full access to cached content in Yemen
Background sync	Background Sync API	Diaspora submissions queue when offline
Push notifications	Firebase Cloud Messaging	Heritage news & updates to installed users
App shortcuts	manifest.json shortcuts[]	4 deep links: Map, Students, YBC, Archives
Install prompt	beforeinstallprompt event	Custom banner after 4s + nav button
Offline indicator	navigator.onLine + events	Top bar + auto-redirect when reconnected
Language persist	localStorage(yai-lang)	Remembers AR/EN preference across sessions
Role persist	localStorage(yai-role)	One-time onboarding, all events tagged

SEO & Browser Introduction

The "browser introduction" is the full meta stack that controls how the platform appears in search engines, social media previews, and browser interfaces. YAI implements a comprehensive SEO architecture on every page.

Meta element	Purpose	YAI implementation
<title>	Browser tab + search title	Bilingual EN/AR — 55–65 characters
meta description	Search result snippet	150-160 chars, bilingual-aware content
og:title / og:image	Facebook, LinkedIn, WhatsApp Preview	Custom per-page OG image, 1200×630px
twitter:card	X/Twitter preview card	summary_large_image on all pages
hreflang en/ar	Language targeting	/en/ and /ar/ alternate versions
rel=canonical	Duplicate content prevention	Explicit canonical on every page
JSON-LD WebSite	Google site identity	SearchAction for sitelinks search box
JSON-LD CulturalOrg	Entity recognition	Type, name, knowsAbout, sameAs
JSON-LD FAQPage	Featured snippet eligibility	2 key questions on homepage
apple-mobile-web-app-*	iOS PWA experience	Black-translucent status bar, short name YAI
msapplication-*	Windows tile	Custom tile image + navy background

Deployment Guide

Step	Command / Action	Notes
1. Clone	<code>git clone https://github.com/your-org/yai.git</code>	Fork or clone the repository
2. Local test	<code>npm serve . -p 3000</code>	Test PWA with service worker locally
3. Lighthouse	<code>npm lighthouse http://localhost:3000 --view</code>	Verify 100 PWA score before deploy
4. Vercel install	<code>npm i -g vercel</code>	Install Vercel CLI
5. Deploy	<code>vercel deploy --prod</code>	Deploys to Vercel global edge
6. Custom domain	domains add yai.org	Add custom domain in Vercel dashboard
7. Analytics	Add domain to Plausible dashboard	plausible.io → Add site → yai.org
8. Push noti	Configure Firebase project	Add FCM VAPID key to sw.js
9. GitHub Actions	Add VERCEL_TOKEN secret	Settings → Secrets → VERCEL_TOKEN

Global Partners

Organisation	Country	Role in YAI	Web
UNESCO	International	World Heritage alignment — Sana'a, Shibam, Zabid, Shabwa	unesco.org
ALIPH	International	Post-conflict protocol — YBC-H07 alignment	aliph.org
CEFAS	France / Yemen	Archaeological research, Chroniques Yéménites	cefas.ye
DAI	Germany	Ma'rib excavations, Sabaeen Kingdom research	dainst.org
MAIRY	Italy	Italian Archaeological Mission — Baraqish, Timriya, Qataban	mairy.it
AIYS	USA / Yemen	American Institute for Yemeni Studies, Yemen Update journal	aiys.org
British-Yemeni Society	UK	Annual journal, bibliography, UK network	b-ys.org.uk
Iconem	France	3D photogrammetric documentation of heritage sites	iconem.com

Project Roadmap

Phase	Timeline	Deliverables	Status
Phase 1 Foundation	Q2–Q3 2026	Homepage + interactive map + YBC sector + Archives sector Design system finalised · Bilingual AR/EN · PWA shell	In progress
Phase 2 Content	Q3–Q4 2026	Students sector · Details sector · General sector All 23 character cards fully documented Diaspora portal launched	Planned
Phase 3 Code	Q4 2026	All 8 YBC chapters — peer review by 3 Yemen-specialist architects YBC PDF exports for all chapters Arabic translation of all code chapters	Planned
Phase 4 Documentation	Q1 2027	Video oral history archive — 10+ master builders documented 3D model integration (Iconem partnership) School curriculum materials in Arabic	Planned
Phase 5 Global	Q2 2027	App Store + Play Store distribution (Capacitor wrapper) French and Italian UI translations Academic citation integration Annual design competition launched	Future

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